July 2, 2025

Terumo Cardiovascular Systems THI Plant
Rahul Reddy Chinkeri
Senior Regulatory Affairs Specialist
6190 Jackson Rd
Ann Arbor, Michigan 48103

Re: K250684
Trade/Device Name: OPUSWAVE Dual Sensor Imaging System
Regulation Number: 21 CFR 892.1560
Regulation Name: Ultrasonic Pulsed Echo Imaging System
Regulatory Class: Class II
Product Code: NQQ, IYO
Dated: March 6, 2025
Received: June 3, 2025

Dear Rahul Reddy Chinkeri:

We have reviewed your section 510(k) premarket notification of intent to market the device referenced above and have determined the device is substantially equivalent (for the indications for use stated in the enclosure) to legally marketed predicate devices marketed in interstate commerce prior to May 28, 1976, the enactment date of the Medical Device Amendments, or to devices that have been reclassified in accordance with the provisions of the Federal Food, Drug, and Cosmetic Act (the Act) that do not require approval of a premarket approval application (PMA). You may, therefore, market the device, subject to the general controls provisions of the Act. Although this letter refers to your product as a device, please be aware that some cleared products may instead be combination products. The 510(k) Premarket Notification Database available at https://www.accessdata.fda.gov/scripts/cdrh/cfdocs/cfpmn/pmn.cfm identifies combination product submissions. The general controls provisions of the Act include requirements for annual registration, listing of devices, good manufacturing practice, labeling, and prohibitions against misbranding and adulteration. Please note: CDRH does not evaluate information related to contract liability warranties. We remind you, however, that device labeling must be truthful and not misleading.

If your device is classified (see above) into either class II (Special Controls) or class III (PMA), it may be subject to additional controls. Existing major regulations affecting your device can be found in the Code of Federal Regulations, Title 21, Parts 800 to 898. In addition, FDA may publish further announcements concerning your device in the Federal Register.

Additional information about changes that may require a new premarket notification are provided in the FDA guidance documents entitled "Deciding When to Submit a 510(k) for a Change to an Existing Device"

(https://www.fda.gov/media/99812/download) and "Deciding When to Submit a 510(k) for a Software Change to an Existing Device" (https://www.fda.gov/media/99785/download).

Your device is also subject to, among other requirements, the Quality System (QS) regulation (21 CFR Part 820), which includes, but is not limited to, 21 CFR 820.30, Design controls; 21 CFR 820.90, Nonconforming product; and 21 CFR 820.100, Corrective and preventive action. Please note that regardless of whether a change requires premarket review, the QS regulation requires device manufacturers to review and approve changes to device design and production (21 CFR 820.30 and 21 CFR 820.70) and document changes and approvals in the device master record (21 CFR 820.181).

Please be advised that FDA's issuance of a substantial equivalence determination does not mean that FDA has made a determination that your device complies with other requirements of the Act or any Federal statutes and regulations administered by other Federal agencies. You must comply with all the Act's requirements, including, but not limited to: registration and listing (21 CFR Part 807); labeling (21 CFR Part 801); medical device reporting (reporting of medical device-related adverse events) (21 CFR Part 803) for devices or postmarketing safety reporting (21 CFR Part 4, Subpart B) for combination products (see https://www.fda.gov/combination-products/guidance-regulatory-information/postmarketing-safety-reporting-combination-products); good manufacturing practice requirements as set forth in the quality systems (QS) regulation (21 CFR Part 820) for devices or current good manufacturing practices (21 CFR Part 4, Subpart A) for combination products; and, if applicable, the electronic product radiation control provisions (Sections 531-542 of the Act); 21 CFR Parts 1000-1050.

All medical devices, including Class I and unclassified devices and combination product device constituent parts are required to be in compliance with the final Unique Device Identification System rule ("UDI Rule"). The UDI Rule requires, among other things, that a device bear a unique device identifier (UDI) on its label and package (21 CFR 801.20(a)) unless an exception or alternative applies (21 CFR 801.20(b)) and that the dates on the device label be formatted in accordance with 21 CFR 801.18. The UDI Rule (21 CFR 830.300(a) and 830.320(b)) also requires that certain information be submitted to the Global Unique Device Identification Database (GUDID) (21 CFR Part 830 Subpart E). For additional information on these requirements, please see the UDI System webpage at https://www.fda.gov/medical-devices/device-advice-comprehensive-regulatory-assistance/unique-device-identification-system-udi-system.

Also, please note the regulation entitled, "Misbranding by reference to premarket notification" (21 CFR 807.97). For questions regarding the reporting of adverse events under the MDR regulation (21 CFR Part 803), please go to https://www.fda.gov/medical-devices/medical-device-safety/medical-device-reporting-mdr-how-report-medical-device-problems.

For comprehensive regulatory information about medical devices and radiation-emitting products, including information about labeling regulations, please see Device Advice (https://www.fda.gov/medical-devices/device-advice-comprehensive-regulatory-assistance) and CDRH Learn (https://www.fda.gov/training-and-continuing-education/cdrh-learn). Additionally, you may contact the Division of Industry and Consumer Education (DICE) to ask a question about a specific regulatory topic. See the DICE website (https://www.fda.gov/medical-devices/device-advice-comprehensive-regulatory-assistance/contact-us-division-industry-and-consumer-education-dice) for more information or contact DICE by email (DICE@fda.hhs.gov) or phone (1-800-638-2041 or 301-796-7100).

Sincerely,

Aneesh S. Deoras -S

Aneesh Deoras
Assistant Director
Division of Cardiac Electrophysiology,
Diagnostics, and Monitoring Devices
Office of Cardiovascular Devices
Office of Product Evaluation and Quality
Center for Devices and Radiological Health

Enclosure

Indications for Use

Please type in the marketing application/submission number, if it is known. This textbox will be left blank for original applications/submissions.	K250684

Please provide the device trade name(s).

OPUSWAVE Dual Sensor Imaging System

Please provide your Indications for Use below.

The OPUSWAVE System with DualView Catheter is intended for the intravascular imaging of coronary arteries and is indicated in patients who are candidates for transluminal interventional procedures

Please select the types of uses (select one or both, as applicable).

- ☒ Prescription Use (Part 21 CFR 801 Subpart D)
- ☐ Over-The-Counter Use (21 CFR 801 Subpart C)

OPUSWAVE Dual Sensor Imaging System
510(k) Summary

This section is prepared in accordance with 21 CFR Part 807.92

I. General Information

Submitters Name and Address:	Terumo Cardiovascular Systems THI Plant 6190 Jackson Road Ann Arbor, MI 48103
Name of the Contact Person:	Rahul Reddy Chinkeri, Senior RA Specialist
Phone Number:	Tel: +1 734-822-1877
Email:	rahul.chinkeri@terumomedical.com
Establishment Registration:	3005228872
Date Prepared:	28 February 2025

II. Device Information

Trade Name:	OPUSWAVE Dual Sensor Imaging System
Common/Generic Name:	Optical Coherence Tomography (OCT) Imaging System / Ultrasonic pulsed echo imaging system
Classification Name:	Ultrasonic pulsed echo imaging system
Device Class:	Class II
Classification Panel:	Radiology/Cardiology Panel
Regulation Number:	21 CFR 892.1560
Product Code(s):	NQQ / IYO

III. Predicate Device

Predicate Device#1, iLab Ultrasound Imaging System cleared under K152316.

Predicate Device#2, Ilumien Optis cleared under K141453.

IV. Device Description

The OPUSWAVE Dual Sensor Imaging System consists of a wheeled console with monitor, keyboard, mouse, a software graphical user interface and a Motor Drive Unit (MDU) protected by an MDU cover. The MDU is connected to a DualView Catheter capable of imaging both, Optical Coherence Tomography (OCT) and Intravascular Ultrasound (IVUS) modalities, either simultaneously or asynchronously, without removing the catheter from the imaging site. The system allows image data to be exported and stored on external media (USB, DVD), and for integration with Cath Lab imaging technologies (angio, ECG).

The sterile operator (physician) is able to control image acquisition by manually positioning the imaging sensor as well as performing pullback (automatically or manually) for defined regions of interest. The system provides analysis tools such as area and linear measurements.

V. Indications for Use

The OPUSWAVE System with DualView Catheter is intended for the intravascular imaging of coronary arteries and is indicated in patients who are candidates for transluminal interventional procedures.

VI. Comparison of Intended Use / Indications to the Predicate Device

The subject and the predicate devices have the same intended use as the devices are intended to provide intravascular imaging of coronary arteries and is indicated in patients who are candidates for transluminal interventional procedures.

VII. Comparison of Technological Characteristics

Attribute	Proposed Device #1 OPUSWAVE Dual Sensor Imaging System	Predicate Device #1 iLab Ultrasound Imaging System	Predicate Device #2 Ilumien Optis	Discussion
Manufacturer	Terumo	Boston Scientific	Abbott	-
Trade Name	OPUSWAVE Dual Sensor Imaging System	iLab Ultrasound Imaging System	Ilumien Optis	-
Common Name	Optical Coherence Tomography (OCT) Imaging System / Ultrasonic pulsed echo imaging system	Ultrasonic pulsed echo imaging system	Optical Coherence Tomography (OCT) Imaging System	Equivalent
Regulation Number	21 CFR 892.1560	21 CFR 892.1560	21 CFR 892.1560	Equivalent
Regulation Name	Ultrasonic pulsed echo imaging system	Ultrasonic pulsed echo imaging system	Ultrasonic pulsed echo imaging system	Equivalent
Regulation Class	II	II	II	Equivalent
510(k) Clearance Number	Subject of this submission	K152316	K141453	-
Product Code	NQQ / IYO	IYO	NQQ	Equivalent
Form factor	Cart-based Console	Cart-based Console	Cart-based Console	Equivalent
Catheter Mechanical Drive	Motor Drive Unit (MDU)	Motor Drive Unit (MDU)	Drive-motor and Optical Controller (DOC)	Equivalent
User Interface	PC-based system with Keyboard and Mouse	PC-based system with Touchpad and Mouse	PC-based system with Keyboard and Mouse	Equivalent
Software Interface	Custom GUI	Custom GUI	Custom GUI	Equivalent

Attribute	Proposed Device #1 OPUSWAVE Dual Sensor Imaging System	Predicate Device #1 iLab Ultrasound Imaging System	Predicate Device #2 Ilumien Optis	Discussion
Image Output	Real-time grayscale image	Real-time grayscale image	Real-time grayscale image	Equivalent
File archive	Network, USB, Optical, Hard drive	Network, Optical, Hard drive	Network, USB, Optical, Hard drive	Equivalent
IVUS Imaging Mode	B Mode	B Mode	Not applicable	Equivalent to predicate 1
IVUS center frequency	40MHz	40MHz	Not applicable	Equivalent to predicate 1
Acoustic Output	Does not exceed Track 3 limits	Does not exceed Track 1 limits.	Not applicable	Equivalent to predicate 1: Dual Sensor will meet 60601-2-37 req.
OCT laser wavelength	1300 nm	Not applicable	1305 nm	Equivalent to predicate 2
Scanning Laser Source Output	Class 1 Laser Output per 60825-1	Not applicable	Class 1M Laser Output per 60825-1	Equivalent to predicate 2
Coronary flush required (OCT imaging only)	Yes; Saline or Contrast or Saline/Contrast mixture	Not required	Yes	Equivalent – coronary flush is required for OCT imaging modes
Pullback Trigger	Automatic or Manual	Automatic or Manual	Automatic or Manual	Equivalent

Table 1: Comparison of Technological Characteristics between the OPUSWAVE Dual Sensor Imaging System *(Subject Device), Lab Ultrasound Imaging System (Predicate Device#1) and Ilumien Optis (Predicate Device#2)*

VIII. Performance Data

The following performance data were provided in support of the substantial equivalence determination:

Electrical Safety and Electromagnetic Compatibility (EMC)

Evaluation for electrical safety and EMC testing was conducted on the OPUSWAVE Dual Sensor Imaging System. The OPUSWAVE Dual Sensor Imaging System complies with the IEC 60601-1 standard for electrical safety and the IEC 60601-1-2 standard for EMC.

Software Verification and Validation Testing

Software verification and validation testing was completed successfully. The fulfillment documentation is provided as recommended by FDA's Guidance for Industry and FDA Staff, "Guidance for the Content of Premarket Submissions for Software Contained in Medical Devices". The software documentation level required for OPUSWAVE Dual Sensor Imaging System is considered as "Enhanced" level since a failure of the software to perform the functions correctly could result in serious injury or death.

Design Verification and Validation Testing

Design verification testing was conducted and demonstrates that OPUSWAVE Dual Sensor Imaging System performs pursuant to the defined design input requirements for the subject device. Design validation, including simulated use testing, was conducted and demonstrates that the OPUSWAVE Dual Sensor Imaging System meets user needs and intended use.

Animal Study

Animal testing was performed and demonstrated the substantial equivalence of the OPUSWAVE Dual Sensor Imaging System to the predicate devices and is included as part of this premarket notification.

Clinical Studies

Clinical testing was not required to demonstrate the substantial equivalence of the OPUSWAVE Dual Sensor Imaging System to the predicate devices and is not included as part of this premarket notification.

IX. Conclusion

The information provided above supports that the OPUSWAVE Dual Sensor Imaging System is substantially equivalent to the predicate devices with respect to the intended use and technological characteristics. Verification and validation testing supports that the hardware and software difference do not raise any new issues of safety and effectiveness. Therefore, it is concluded that the OPUSWAVE Dual Sensor Imaging System is substantially equivalent to the predicate device.